MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

GATE, EXCLUSIVE OR, DIGITAL, KEY CODE BMC

This amendment forms a part of Military Specification MIL-M-28787/255(EC), dated 1 June 1982, and is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

PAGES 4, 5, and 6

TABLE I: Delete and substitute new table I as printed on pages 2, 3, and 4 of this amendment.

PAGE 7

4.5.1, ninth line: Delete "100°C" and substitute "85°C".

PAGE 8

6.4, second line: Delete "100°C" and substitute "85°C".

6.4, Output drive current (V_{DD} = 5 V): Delete "100°C" (two places) and substitute "85°C" (two places).

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FIGURE 8, NOTE 2: Delete "100°C" and substitute "85°C".

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FIGURE 9, NOTE 2: Delete "100°C" and substitute "85°C".

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5963-N497)

FSC 5963

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation Initial	III	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	III	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B1-B24	Input clamp diode voltage, negative	IV	X	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	V
C1-C24	Input clamp diode voltage, positive	V	X	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	V
D1-D12	Static output voltage, low level (I_{OL}= 0)	VI	X	0.00	0.05	0.00	0.05	0.00	0.05	0.00	0.05	0.00	0.05	V
E1-E12	Static output voltage, high level (I_{OH} = 0)	VII	X	4.95	5.05	4.95	5.05	4.95	5.05	4.95	5.05	4.95	5.05	V
F1-F24	Output voltage, low level, V_{DD} = 5V	VIII	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
G1-G24	Output voltage, high level, V_{DD} = 5V	IX	X	4.6	5.0	4.6	5.0	4.6	5.0	4.6	5.0	4.6	5.0	V
H1-H12	Output voltage, low level, V_{DD} = 15V	X	X	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	V
J1-J12	Output voltage, high level, V_{DD} = 15V	XI	X	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	V
K1-K24	Input current, low level	XII	X	0	-200	---	---	0	-200	---	---	0	-200	nA
L1-L24	Input current, high level	XIII	X	0	200	---	---	0	200	---	---	0	200	nA
M1	Power supply current, low level	XIV	X	0	6	0	6	0	60	0	6	0	98	μA
M2	Power supply current, high level	XIV	X	0	6	0	6	0	60	0	6	0	98	μA
N1-N12	Transition time (t_{TLH}), C_L= 50 pF, V_{DD}= 5V	XV	S	20	200	20	200	22	220	15	150	23	236	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
M13-M24	Transition time (t_{TLH}), C_L = 150 pF, V_{DD} = 5V	XV	Q	68	680	68	680	75	750	51	5115	80	803	ns
P1-P12	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 5V	XVI	S	20	200	20	200	22	220	15	150	23	236	ns
P13-P24	Transition time (t_{THL}), C_L = 150 pF, V_{DD} = 5V	XVI	Q	68	680	68	680	75	750	51	515	80	803	ns
Q1-Q12	Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 10V	XVII	Q	10	100	10	100	11	110	7	75	12	119	ns
R1-R12	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 10V	XVIII	Q	10	100	10	100	11	110	7	75	12	119	ns
S1-S12	Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 15V	XIX	Q	8	80	8	80	9	90	6	60	10	96	ns
T1-T12	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 15V	XX	Q	8	80	8	80	9	90	6	60	10	96	ns
U1-U24	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 5V	XXI	S	35	350	35	350	38	385	26	265	41	413	ns
U25-U48	Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5V	XXI	Q	55	550	55	550	61	610	42	420	65	651	ns
V1-V24	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5V	XXII	S	35	350	35	350	38	385	26	265	41	413	ns
V25-V48	Propagation delay time (t_{PHL}), C_L = 150 pF, V_{DD} = 5V	XXII	Q	55	550	55	550	61	610	42	420	65	651	ns
W1-W24	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 10V	XXIII	Q	15	150	15	150	16	165	11	115	17	178	ns
X1-X24	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10V	XXIV	Q	15	150	15	150	16	165	11	115	17	178	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
Y1-Y24	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 15V	XXV	Q	10	100	10	100	11	110	7	75	12	119	ns
Z1-Z24	Propagation delay time, (t_{PHL}), C_L = 50 pF, V_{DD} = 15V	XXVI	Q	10	100	10	100	11	110	7	75	12	119	ns
AA	Power supply filter capacitance	XXVII	S	0.007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	µF
BB1, BB2	C2 and C3 capacitance	XXVIII	X	176	264	176	264	176	264	176	264	176	264	pF
CC1-CC24	Input capacitance	XXIX	Q	2	15	2	15	2	15	2	15	2	15	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified, limits are for initial and end-of-life.

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